

Patterns

(Class Slides)

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Note: We are sharing these slides as lot of mathematical logic building was done on paper.

Hope these will help you in your learning process 

1 1 1 1
2 2 2 2
3 3 3 3
4 4 4 4

*
* *
* * *
* * * *

* * * *
* * *
* *
*

1
12
123
1234

A
BC
DEF
GHIJ

* * * * *
* * * * *
* * * * *
* * * * *

*
* *
* * *
* * * *

1
2 3
4 5 6
7 8 9 10
11 12 13 14 15

*
* * *
* * * * *
* * * * * * *
* * * * * * *
* * * * *
* * *
*

* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *

Nested Loops

Loop inside a loop

	c1	c2	c3	c4
R1	1	1	1	1
R2	2	2	2	2
R3	3	3	3	3
R4	4	4	4	4

$n = 4$

n

① Outer loop : no. of Rows

② Inner loop : no. of columns /
each row

③ Work in inner loop (each row)
i print

```
for (int i=1; i<=4; i++) {
```

```
    for (int j=1; j<=4; j++) {
```

```
        cout << i << " " << j << " ";
```

```
    }
```

```
}
```

Print Star pattern

n = 4

R1 * 1st
R2 ** 2st
R3 *** 3st
R4 **** 4st

```
for (int i = 1; i <= n; i++) {  
    for (int j = 1; j <= i; j++) {  
        cout << "*" << " ";  
    }  
    cout << endl;  
}
```

n = 4 }

① outer loop → Rows (i) → Row no.
n times (1 to n)

② inner loop (each row's)
columns
i times (1 to i)

③ work ?

cout << "*" << " "

Print Inverted Star pattern

$n = 4$

<u>R1</u>	* * * *	$4 = n - i + 1$
<u>R2</u>	* * *	$3 = n - i + 1$
<u>R3</u>	* *	$2 = n - i + 1$
<u>R4</u>	*	$1 = n - i + 1$

① outer loop (rows)
(1 to n)

② inner loop (each row)
(1 to $n - i + 1$)

③ work?
cout << "*" << endl

n = 4

Print Half Pyramid pattern

$n=4$

R1	1	1
R2	12	1 to 2
R3	123	1 to 3
R4	1234	1 to 4

1 to i

```
for (int i = 1; i <= n; i++) {  
    for (int j = 1; j <= i; j++) {  
        cout << j << " ";  
    }  
    cout << endl;  
}
```

$n = 4$

① outer loop (rows)
(1 to n)

② inner loop (each row)
(1 to i) $\Rightarrow j$

③ work?
 $cout << j << " ";$

Print Character Pyramid pattern

A
BC
DEF
GHIJ

$n = 4$

char ch = 'A'

① outer loop (row)
(1 to n)

② inner loop (each row)
(1 to i)

③ work?

cout << ch;
ch++;
||

$n = 4$

Print Hollow Rectangle pattern

<u>R1</u>	*	*	*	*	*	1st + 3st + 1st
R2	*	-	-	-	*	1st + 3sp + 1st
R3	*	-	-	-	*	1st + 3sp + 1st
<u>R4</u>	*	*	*	*	*	1st + 3st + 1st

first or last $\rightarrow$ 5 stars
(1) (n)

① Outer loop (rows)
(1 to n)

② Inner loop (each row)

cout << "*" ; // First

for (1 to n-1) {
1st or last $\rightarrow$ "*" } ② work

else $\rightarrow$ "
cout << "*" ; // last

n = 4

$n=4$

Inverted & Rotated Half-Pyramid

— — — *	R1	3 sp	+ 1 st
— — **	R2	2 sp	+ 2 st
— ***	R3	1 sp	+ 3 st
****	R4	0 sp	+ 4 st

$n - i$ spaces

$i=1$

$4-1 = 3$

$i=2$

$4-2 = 2$

$i=3$

$4-3 = 1$

$i=4$

$4-4 = 0$

$n = 4$

① outer loop (rows)
(1 to n)

② inner loop (each row)

a) Spaces (1 to $n-i$)
`cout << " "` ← work

b) Stars (1 to i)
`cout << "*" ← work`

`cout << endl`

Print Floyd's Triangle

num = 1

R1	1									
R2	2	3								
R3	4	5	6							
R4	7	8	9	10						
R5	11	12	13	14	15					

ith i times
(1 to i)

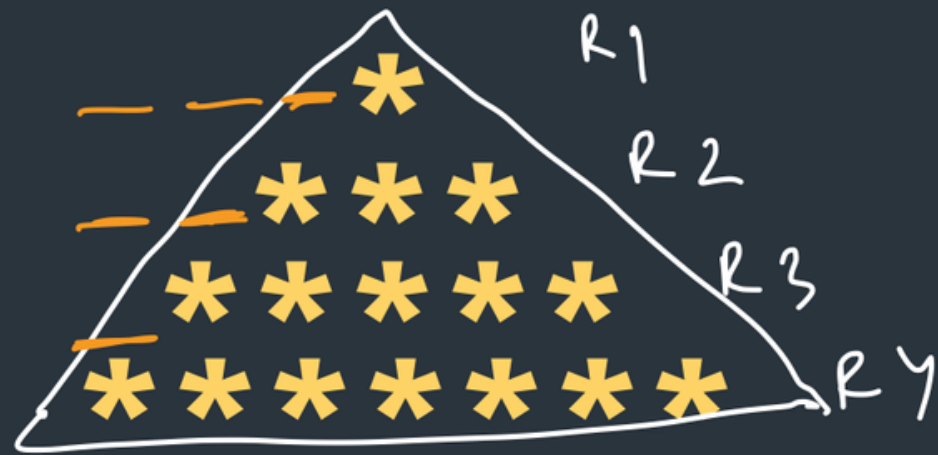
① outer loop (rows)
(1 to n)

② inner loop (each row)
elements
(1 to i)

③ work?
cout << num;
num++;

Diamond Pattern

n = 4



3sp + 1st
2sp + 3st
1sp + 5st
0sp + 7st

1st pyramid

① outer loop (rows)
(1 to n)

② inner loop (each row)
elements

a) (1 to n-i) cout << " "

b) (1 to 2*i-1) cout << "*" << " "

i = 1	→	1	$2 \times i - 1 = 1$
i = 2	→	3	$2 \times i - 1 = 3$
i = 3	→	5	$2 \times i - 1 = 5$
i = 4	→	7	$2 \times i - 1 = 7$

n = 4

Diamond Pattern

$n = 4$

* * * * * R₄
- * * * * R₃
- - * * * R₂
- - - * R₁

0 sp + 7 st
1 sp + 5 st
2 sp + 3 st
3 sp + 1 st
 $2 \times i - 1$

2nd pyramid

① outer loop (rows)
(n to 1)

② inner loops (each row elements)

a) sp (1 to $n - i$)

b) st (1 to $2 \times i - 1$)

$n = 4$

Print Butterfly Pattern

n=4

```

*
**
***
****
****
***
**
*

```

```

*   *
**  **
*** ***
****
****
***  **
**   **
*    *

```

R1 : 1st + 6sp + 1st
 R2 : 2st + 4sp + 2st
 R3 : 3st + 2sp + 3st
 R4 : 4st + 0sp + 4st

$2 * (n - i)$

Pattern1

① outer loop (rows)
(1 to n)

② inner loop

a) stars (1 to i)

b) spaces (1 to $2 * (n - i)$)

c) stars (1 to i)

n = 4

Print Butterfly Pattern

n=4



R4 4 st
R3 3 st
R2 2 st
R1 1 st

✓
0 sp 4 st
2 sp ✓ 3 st
4 sp ✓ 2 st
6 sp ✓ 1 st

$$2 * (n - i) = 2 * (4 - 1) = 6$$

i = 1

Pattern 2
① outer loop (rows)
(n to 1)

② inner loop
a) stars (1 to i)

b) spaces (1 to $2 * (n - i)$)

c) stars (1 to i)

n = 4